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DEVICES HAVING POROUS STRUCTURES AND METHODS FOR USING THE SAME

Abstract

The present disclosure includes fixation devices, such as an orthopedic screw or implant, that comprises one or more porous elements or fenestrations to aid in osteo-integration of the fixation device. The fixation device may be additively manufactured using biocompatible materials such that the solid and porous aspects of the screw are fused together into a single construct. In yet another aspect, the fixation device comprises at least a portion or section incorporating a porous structure, which enables bony ingrowth through the porous section/portion of the screw, and thereby facilitates biocompatibility and improve mechanical characteristics. Methods for using the fixation device are also described herein.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/535,123, filed Aug. 29, 2023. This application is also a continuation-in-part of U.S. patent application Ser. No. 17/807,823, filed on Jun. 20, 2022, now issued as U.S. Pat. No. 11,925,400, which in turn is continuation of U.S. patent application Ser. No. 17/008,819, filed Sep. 1, 2020, now U.S. Pat. No. 11,376,049, which is a continuation of U.S. patent application Ser. No. 15/675,104, filed Aug. 11, 2017, now U.S. Pat. No. 10,758,283, which in turn is continuation-in-part of U.S. patent application Ser. No. 15/416,975, filed on Jan. 26, 2017, now U.S. Pat. No. 9,987,024, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/373,855, filed Aug. 11, 2016, the entirety of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present disclosure relates to the field of medical devices generally. More specifically, the present disclosure relates to a fixation device, such as a screw, comprising one or more porous portions and/or fenestrations. Systems and methods for using the foregoing devices are also disclosed herein.

BACKGROUND

[0003] Individuals who suffer degenerative disc disease, natural spine deformations, a herniated disc, spine injuries or other spine disorders often require surgery on the affected region to relieve pain and prevent further injury. Such spinal surgeries may involve fixation of two or more adjacent vertebral bodies. For patients with varying degrees of degenerative disc disease and/or nerve compression with associated lower back pain, spinal fusion surgery or lumbar arthrodesis (“fusion”) is commonly used to treat the degenerative disease. Fusion commonly involves distracting and/or decompressing one or more intervertebral spaces, followed by removing any associated facet joints or discs, and then joining or “fusing” two or more adjacent vertebra together. Fusion of vertebral bodies also commonly involves fixation of two or more adjacent vertebrae, which may be accomplished through introduction of rods or plates, and screws or other devices into a vertebral joint to join various portions of a vertebra to a corresponding portion on an adjacent vertebra. Given the complexities of surgical procedures, as well as anatomical variation between patients who receive surgical devices, it is often challenging to provide a device or implant that achieves the needs of a particular patient without completely customizing the device or implant for a single patient.

[0004] Many prior art fixation devices suffer from significant disadvantages, such as poor stability, poor flexibility, poor accuracy, difficulty in handling, lack of customized features, inability to combine with other materials, loss of fixation over time, subsidence and other disadvantages.

Certain fixation devices also impair visibility and provide little or no ability for the operator to gauge depth or accuracy. These problems and shortcomings are even more noticeable for fixation devices used in surgical settings or which otherwise require precision.

[0005] In addition, fixation devices used in surgical settings can also suffer from further shortcomings. For example, pedicle screws are subject to relatively high failure rates, which is often attributed to a failure of the bone-screw interface. Screws for use in surgical settings may also be limited for use in only certain bony anatomies, or with only certain types of drilling apparatus, and may not be suitable for combination with other devices or materials.

[0006] Accordingly, there is a need for a fixation device that decreases the mean time for affixing the device to the desired location, enhances depth control, stability and accuracy, and which otherwise overcomes the disadvantages of the prior art. There is also need for a more customized fixation device, such as an orthopedic screw, which includes one or more porous elements or fenestrations to aid in osteo-integration when implanting the fixation device. The fixation device may be additively manufactured using biocompatible materials such that the solid and porous aspects of the device are fused together into a single solid construct, and potentially having the porous elements interdigitated within and around various solid elements of the device.

[0007] The prior art also fails to teach a system for creating a customized fixation device based on patient data, such as data derived from a patient's MRI or CT scan. For example, the availability of patient-specific data (for example, a vertebral body) may allow a surgeon to accommodate for subtle variations in the position and orientation of a screw or other fixation device to avoid particular bony anatomy, or irregularities in the positioning and alignment of the adjoining vertebral bodies. As another example, the use of patient data may also assist a surgeon in selecting a desired trajectory for an fixation device so as to avoid, for example, crossing the pedicle wall and violating the spinal canal during a spine-related procedure. The use of patient-specific data permits the surgeon to avoid these types of mistakes and may comprise specific orientation, end-stops/hard stops, or other safety related features to avoid over-torque or over-insertion of the fixation device. This data also permits the surgeon to quickly and efficiently locate and place devices with corresponding patient-contacting surface(s), while ensuring the fixation device is in the appropriate location and orientation.

[0008] It would therefore be advantageous to provide a fixation device that significantly reduces, if not eliminates, the shortcoming, problems and risks noted above. Other advantages over the prior art will become known upon review of the Summary and Detailed Description and the appended claims.

SUMMARY

[0009] According to various embodiments presented herein, the present disclosure describes a fixation device, such as a screw, comprising one or more fenestrations which permit introduction of at least one other material or substance to through the fenestrations in the fixation device. In other embodiments, the fenestrations permit the fixation device to capture and retain material. In yet another aspect of the present disclosure, a method of using the fixation devices described herein is disclosed, including but not limited to in a surgical setting.

[0010] One particular aspect of the present disclosure involves a fixation device, such as an orthopedic pedicle screw or implant, that is manufactured such that it includes one or more porous elements or fenestrations in order to aid in osteo-integration of the implanted fixation device. For instance, a pedicle screw (as one type of fixation device) may be additively manufactured using biocompatible materials such that the solid and porous aspects of the screw are fused together into a single solid construct with the porous elements interdigitated within and around various solid elements of the screw.

[0011] In yet another aspect, a surgical screw design having at least a portion or section incorporating a porous structure enables bony ingrowth through the porous section/portion of the screw, and thereby facilitate biocompatibility and improve mechanical characteristics. The porous

elements of the screw may be designed to more closely resemble that of the patient's anatomy, in order to reduce discontinuities and stress risers at the bone/screw interface. Bony ingrowth within one or more porous elements of the screw in turn facilitates screw pullout strength, and may reduce the risk of loosening of a fixation device under dynamic loading situations.

[0012] In one aspect, the fixation device comprises a porosity gradient. In embodiments, the length, diameter, depth, and/or density of porosity along the gradient is selected according to the properties of adjacent patient bone, which may be derived from MRI, CT, bone density, medical imaging or other patient-specific data.

[0013] In another aspect, the gradient of porosity is primarily along a single axis of the fixation device, such as the length of the fixation device. In one embodiment, the gradient is oriented such that the distal end of the fixation device has a greater percentage of fenestrations than the proximal end.

[0014] In yet another aspect, the gradient is selected according to patient imaging data, including but not limited to a computed tomography scan, x-ray imaging, bi-planar x-ray imaging, magnetic resonance imaging, and bone densitometry scan.

[0015] In yet another aspect, the fixation device may be coupled with a washer comprising one or more spikes or teeth. In embodiments, the washer is configured to be advanced along a shaft of the fixation device and interface with a patient's boney anatomy, such as during implantation of the fixation device.

[0016] In yet another aspect, the shaft of the fixation device comprises an internal channel, which may be oriented along substantially the entire length of the shaft. In embodiments, the internal channel is accessible from the proximal end of the device, wherein the internal channel is further configured to receive graft or equivalent material.

[0017] In yet another aspect, the porous gradient is present on about 10-30 percent of the shaft of the fixation device. In another embodiment, the porous gradient is present on about 30-60 percent of the shaft of the fixation device.

[0018] Additional aspects of the present disclosure are provided in the Appendix hereto, which is incorporated by reference in its entirety.

[0019] Incorporated by reference in their entireties are the following U.S. patents and patent applications directed generally to methods and apparatus related to surgical procedures, thus providing written description support for various aspects of the present disclosure. The U.S. patents and pending applications incorporated by reference are as follows: U.S. Pat. Nos. 11,376,049, 10,758,283, 7,957,824, 7,844,356, 7,658,610, 6,830,570, 6,368,325, 3,486,505 and U.S. Pat. Pub. Nos. 2010/0217336, 2009/0138020, 2009/0087276, 2008/0161817, 2008/0114370, and 2007/0270875.

[0020] Additionally, U.S. Pat. Nos. 8,758,357, 8,870,889, 9,198,678 and 9,642,633 are incorporated by reference for the express purpose of illustrating systems and methods for creating an device, such as the one described herein, using additive manufacturing or other techniques, wherein the device incorporates one or more patient-matched surfaces or is otherwise customized to a particular patient.

[0021] The phrases “at least one,” “one or more,” and “and/or,” as used herein, are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C,” “at least one of A, B, or C,” “one or more of A, B, and C,” “one or more of A, B, or C,” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

[0022] Unless otherwise indicated, all numbers expressing quantities, dimensions, conditions, and so forth used in the specification and claims are to be understood as being approximations which may be modified in all instances as required for a particular application of the novel apparatus described herein.

[0023] The term “a” or “an” entity, as used herein, refers to one or more of that entity. As such, the

terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein.

[0024] The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

Accordingly, the terms “including,” “comprising,” or “having” and variations thereof can be used interchangeably herein.

[0025] It shall be understood that the term “means” as used herein shall be given its broadest possible interpretation in accordance with 35 U.S.C., Section 112(f). Accordingly, a claim incorporating the term “means” shall cover all structures, materials, or acts set forth herein, and all of the equivalents thereof. Further, the structures, materials, or acts and the equivalents thereof shall include all those described in the Summary, Brief Description of the Drawings, Detailed Description, Abstract, and Claims themselves.

[0026] The Summary is neither intended, nor should it be construed, as being representative of the full extent and scope of the present disclosure. Moreover, references made herein to “the present disclosure” or aspects thereof should be understood to mean certain embodiments of the present disclosure, and should not necessarily be construed as limiting all embodiments to a particular description. The present disclosure is set forth in various levels of detail in the Summary as well as in the attached drawings and the Detailed Description, and no limitation as to the scope of the present disclosure is intended by either the inclusion or non-inclusion of elements or components when describing certain embodiments herein. Additional aspects of the present disclosure will become more readily apparent from the Detailed Description, particularly when taken together with the drawings.

[0027] The above-described benefits, embodiments, and/or characterizations are not necessarily complete or exhaustive, and in particular, as to the patentable subject matter disclosed herein. Other benefits, embodiments, and/or characterizations of the present disclosure are possible utilizing, alone or in combination, as set forth above and/or described in the accompanying figures and/or in the description herein below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The accompanying drawings, which are incorporated herein and constitute a part of the specification, illustrate embodiments of the disclosure, and together with the Summary and the Detailed Description serve to explain the principles of these embodiments. In certain instances, details that are not necessary for an understanding of the disclosure or that render other details difficult to perceive may have been omitted. It should be understood, of course, that the present disclosure is not necessarily limited to the particular embodiments illustrated herein. Additionally, it should be understood that the drawings are not necessarily to scale. In the drawings:

[0029] FIG. 1 shows a side elevation view of a fixation device according to one embodiment of the present disclosure;

[0030] FIG. 2 shows a detailed view of the fixation device of FIG. 1;

[0031] FIG. 3 shows a sectional view of a fixation device according to another embodiment of the present disclosure;

[0032] FIG. 4 shows a detailed view of the fixation device of FIG. 3;

[0033] FIG. 5 shows a sectional view of a fixation device according to another embodiment of the present disclosure;

[0034] FIG. 6 shows an elevation view of a fixation device according to another embodiment of the present disclosure;

[0035] FIG. 7 shows a detailed view of the fixation device of FIG. 6;

[0036] FIG. 8 shows a sectional view of a fixation device according to another embodiment of the

present disclosure;

[0037] FIG. **9** shows a perspective, sectional view of the fixation device of FIG. **8**;

[0038] FIG. **10** shows an elevation view of a fixation device according to another embodiment of the present disclosure;

[0039] FIG. **11** shows a sectional view of a fixation device according to yet another embodiment of the present disclosure;

[0040] FIG. **12** shows another sectional view of a fixation device according to yet another embodiment of the present disclosure;

[0041] FIG. **13** shows another sectional view of a fixation device according to yet another embodiment of the present disclosure;

[0042] FIG. **14** shows another sectional view of a fixation device according to yet another embodiment of the present disclosure;

[0043] FIGS. **15A-15B** show detailed views of a fixation device according to another embodiment of the present disclosure;

[0044] FIGS. **16A-16B** show perspective and elevation views of a washer and fixation device according to another embodiment of the present disclosure;

[0045] FIGS. **17A-17C** show perspective views of the fixation device according to the embodiment shown in FIGS. **16A-16B**;

[0046] FIGS. **18A-18B** show perspective and sectional views of a fixation device according to an alternate embodiment;

[0047] FIG. **19** shows a sectional view of a fixation device according to yet another embodiment;

[0048] FIGS. **20A-20B** show perspective views of a fixation device according to yet another embodiment;

[0049] FIGS. **21A-21B** show perspective views of a fixation device according to yet another embodiment; and

[0050] FIG. **22** shows a detailed elevation view of a fixation device according to an alternate embodiment.

[0051] Similar components and/or features may have the same reference number. Components of the same type may be distinguished by a letter following the reference number. If only the reference number is used, the description is applicable to any one of the similar components having the same reference number.

DETAILED DESCRIPTION

[0052] The present disclosure has significant benefits across a broad spectrum of endeavors. It is the Applicant's intent that this specification and the claims appended hereto be accorded a breadth in keeping with the scope and spirit of the disclosure and various embodiments disclosed, despite what might appear to be limiting language imposed by specific examples disclosed in the specifications. To acquaint persons skilled in the pertinent arts most closely related to the present disclosure, preferred and/or exemplary embodiments are described in detail without attempting to describe all of the various forms and modifications in which the novel apparatus, devices, systems and methods might be embodied. As such, the embodiments described herein are illustrative, and as will become apparent to those skilled in the arts, may be modified in numerous ways within the spirit of the disclosure.

[0053] By way of providing additional background, context, and to further satisfy the written description requirements of 35 U.S.C. § 112, the following are incorporated by reference in their entireties for the express purpose of explaining and further describing the various tools and other apparatus commonly associated therewith surgical procedures, including minimally invasive surgery ("MIS") procedures: U.S. Pat. No. 6,309,395 to Smith et al.; U.S. Pat. No. 6,142,998 to Smith et al.; U.S. Pat. No. 7,014,640 to Kemppanien et al.; U.S. Pat. No. 7,406,775 to Funk, et al.; U.S. Pat. No. 7,387,643 to Michelson; U.S. Pat. No. 7,341,590 to Ferree; U.S. Pat. No. 7,288,093 to Michelson; U.S. Pat. No. 7,207,992 to Ritland; U.S. Pat. No. 7,077,864 Byrd III, et al.; U.S. Pat.

No. 7,025,769 to Ferree; U.S. Pat. No. 6,719,795 to Cornwall, et al.; U.S. Pat. No. 6,364,880 to Michelson; U.S. Pat. No. 6,328,738 to Suddaby; U.S. Pat. No. 6,290,724 to Marino; U.S. Pat. No. 6,113,602 to Sand; U.S. Pat. No. 6,030,401 to Marino; U.S. Pat. No. 5,865,846 to Bryan, et al.; U.S. Pat. No. 5,569,246 to Ojima, et al.; U.S. Pat. No. 5,527,312 to Ray; and U.S. Pat. Appl. No. 2008/0255564 to Michelson.

[0054] Referring now to FIGS. **1-22**, varying embodiments of the present disclosure are shown. The fixation devices shown in FIGS. **1-22** preferably comprise a porous or fenestrated structure, which may be exposed at specific regions along the fixation device, and which aid with osteo-integration while increasing the mechanical stability of the device. Various benefits of the fixation device, which in a preferred embodiment is in the form of a screw, are described herein.

[0055] In the embodiment of FIG. **1**, the fixation device is provided in the form of a screw having a head portion **10** and a distal portion **20**, with threads **30** located on the shaft **15** between the head portion **10** and the distal portion **20**. In certain embodiments, the head portion **10** is affixed to the body of the fixation device, while in other embodiments the head is allowed to float and thereby provide a poly-axial screw assembly. In other embodiments, the head may be temporarily attached to the body of the fixation device. According to this particular embodiment, the porous or fenestrated structure is comprised of the shaft **15** of the screw, but not covering the head portion **10** or the threads **30**. The number of fenestrations or porosity of the porous structure of the screw may be greater or lesser than depicted in FIG. **1**.

[0056] Referring now to FIG. **2**, a detailed view of the porous screw is depicted. In this drawing, the fenestrations **40** are arranged in series and extend substantially the entire width of the shaft **15** between the threads **30**. In alternate embodiments, the fenestrations may be fewer in number than shown, and may not be uniform in their location and arrangement. The fenestrations may be circular or any other shape suitable for the fixation device. In alternate embodiments, the fenestrations extend even through the threads **30** of the screw.

[0057] The screw may be additively manufactured such that the porous structure may be exposed to the interfacing bone but also contained within the core of the screw. The screw may be additively manufactured, by way of example but not limitation, out of biocompatible alloys, including by using electron-beam melting or selective laser sintering methods to produce various surface finishes on the porous and solid aspects of the screw. The screw may be manufactured as a single part, fixed angle screw, or may be poly-axial.

[0058] Referring now to FIGS. **3-4**, the screw may comprise a substantially hollow section that is in communication with the plurality of fenestrations **40**. For example, as shown in FIG. **3**, the screw may comprise a porous structure in a solid core to provide superior mechanical characteristics. The porous structure may be confined to only a certain region of the screw, or may extend substantially the entire length of the screw.

[0059] Referring now to FIG. **5**, the fenestrations **40** may extend the entire width of the screw for substantially the entire length of the shaft **15**, as shown. In this manner, the number of fenestrations **40** may be increased, thereby increasing the number of locations along the outer surface of the screw that either material can be introduced (via the screw) or material may be collected (via suction as described above). In embodiments, the screw may have a longitudinal channel **35** that runs lengthwise through the screw and permits material to be introduced through the longitudinal channel and into the plurality of fenestrations **40**. For example, the longitudinal channel may be accessible from the head portion **10** of the screw, such that an operator may inject fluid or solid material into the longitudinal channel **35** and ultimately through the porous structure of the screw via fenestrations **40**. In a similar manner, the fenestrations **40** (which surround the outer surface of the shaft **15** of the screw) may be used with vacuum or suction applied to the head portion **10** of the screw to retract material adjacent to the fenestrations **40** and either retained by the porous structure of the screw or within the longitudinal channel **35**. By way of example, the screw could either be infused with bone morphogenetic protein or equivalent to enhance fusion between the screw and

the patient's anatomy, or the screw could suction and retain autogenous blood into the fenestrations, which in turn stimulates bony ingrowth in the porous section of the screw. Variations on this embodiment are considered within the scope of the present disclosure.

[0060] Referring to FIG. 6, one embodiment comprises a varying array of fenestrations 40, which may increase or decrease along the length of the screw. As shown in FIG. 6, the number of fenestrations 40 may vary in a second portion of the screw, such that there is a decreased number or pattern of fenestrations 40' than at a different portion of the screw. In other portions of the screw, the shaft 15 may not comprise any fenestrations at all. This variation is best shown in the detailed view of FIG. 7. In this manner, material may either be collected or introduced via the screw only at desired locations along the length of the screw, which may be desirable given the particular bone density and surrounding anatomical features of a particular patient. In reference to FIGS. 6 and 7, in certain embodiments the longitudinal channel may comprise additional fenestrations in certain portions to better accommodate a particular patient's anatomy. For instance, a first portion of the longitudinal channel 35 is as depicted in FIG. 3, while a second portion of the longitudinal channel 35' may comprise a greater number of fenestrations and therefore a greater porosity.

[0061] Further variations of the embodiments described above are shown in FIGS. 8-9. For example, the fixation device may be a pedicle screw, which may be additively manufactured such that the solid and porous aspects of the screw are fused together into a single solid construct. In other embodiments, the solid and porous structure may be co-extensive with the porous elements interdigitated within and around various solid elements of the screw. Variations on these embodiments are considered within the spirit of the presently claimed invention.

[0062] The porous elements of the fixation device, screw or implant may be designed to more closely resemble that of a specific patient's anatomy. Accordingly, one advantage of the present disclosure is to promote bony ingrowth throughout the porous portions or sections of the implant, which in turn reduces the risk of loosening of the device under dynamic loading situations.

[0063] In still other embodiments, only a portion of the screw is manufactured with a porous surface. For example, the exposed porous aspects of the screw may be localized along the minor diameter of the thread form. The screw may therefore comprise hollow, porous or solid core elements to allow for varying levels of implant stiffness. These areas may be surrounded by a mostly solid thread form to facilitate smooth implantation and firm seating of the screw. Particular reference is made to FIGS. 3-4 when referring to this embodiment.

[0064] As referred to above, the porous elements of the screw may be designed with localized fenestrations, which may be at least partially accessible from the screw head, and thereby facilitate delivery of osteogenic agents such as bone-morphogenetic proteins, HA and or allograft or autograph tissue into the porous portions of the screw. This in turn allows for bony ingrowth and greater pullout resistance, as described above. The exposed porous structure may be located on the proximal portion of the screw, adjacent the screw head, in order to localize ingrowth. Localization of ingrowth may increase mechanical characteristics of the bone screw interface, and subsequently allow for easier implant removal in the case of revision surgery. The localized porous elements may be tapered outward to increase the interference fit of the porous elements with the surrounding anatomy.

[0065] The porous features are representative of porous cancellous bone with porosity preferably ranging between 30-80% to allow for ingrowth of osteocytes and supporting vasculature. Stated another way, the porous features, when compared to the solid features of the device, make up about 30-80% of the volume of the device. In a most preferred embodiment, the porosity is about 50%.

[0066] In certain embodiments, the porous structure may be regular and geometric or irregular in form. In yet other embodiments, the porous density may be homogenous throughout the screw, or may be heterogeneous in order to attain desired stiffness and or improve the structural interface of the solid and porous elements. Referring now to FIGS. 10-14, various additional embodiments of the present disclosure are shown comprising a porosity gradient about, by way of example and not

limitation, a major dimension of the fixation device.

[0067] A side elevation view of an exemplary porosity gradient along the length of a fixation device, such as a screw, is shown in FIG. **10**. In this particular embodiment, the distal end of the screw **60** has a greater porosity (i.e., a denser concentration of fenestrations **40**) than the proximal end. However, in certain embodiments the fixation device may have a greater porosity towards the central axis of the screw, or vice versa, such that the gradient is oriented primarily along the minor axis of the fixation device. In certain embodiments, including those described below, the fixation device comprises a porosity gradient that varies by more than one axis or dimension of the fixation device.

[0068] In embodiments, the gradient or pattern of porosity present in the fixation device may be derived from, at least in part, the properties of a specific patient who receives the fixation device. For example, the gradient may be determined from properties of a patient's boney anatomy, including data obtained from a bone density scanner or equivalent device. Additional data may be obtained from complementary equipment, including but not limited to magnetic resonance imaging (MRI) data, computed tomography (CT) data, x-ray imaging data, bi-planar x-ray imaging data, bone densitometry scan data, medical imaging data, fluoroscopy data, sampled bone material harvested from the patient, or other anatomical data.

[0069] Referring still to FIG. **10**, the gradient shown comprises a denser porous structure towards the distal end **60**, and preferably comprises a gradual reduction in porosity approaching the midline of the fixation device, as shown. In other embodiments, the porous portions of the fixation device may continue throughout the threaded portion of the screw, or may only continue for a very short length of the screw. The fixation device comprising the porous gradient shown in FIG. **10**, while preferable for spinal surgical procedures and especially pedicle screw trajectories, may also be beneficial in other applications, including but not limited to general orthopedic procedures, S1AI and/or S2AI screw trajectories, spinal tethering anchors, suturing anchors and anchors for ACL repair.

[0070] Referring to FIGS. **11-13**, additional gradients are shown. In these embodiments, the screw preferably comprises a solid or non-porous core **70**, which provides adequate strength for the intended application(s). The screw's core **70**, however, may be tapered and decrease the solid or non-porous diameter of the screw as the gradient of porosity becomes more dense (i.e., approaching the distal end of the screw as shown in FIG. **12**). In other embodiments, this gradient migration is reversed about the major axis of the screw. In other embodiments, and referring to FIG. **13**, the change in diameter of the non-porous core **70** of the screw occurs more gradually over the length of the threaded portion of the screw, and may continue through the entire length of the threaded portion of the screw.

[0071] Attention is now drawn to FIG. **14**. In embodiments, at least a portion of the distal end **60** of the fixation device may be cannulated. This cannulation facilitates preloading the fixation device with autograft, allograft, bioactive or drug eluting material prior to implantation. Alternatively, a user may preload the foregoing material via the pilot hole. In these manners, the fixation device is capable of being preloaded with graft or similar material from within the fixation device, thereby facilitating and promoting boney ingrowth.

[0072] In an alternative embodiment, the cannulation may enable collection of native bone material as the fixation device is being inserted, thereby augmenting or replacing preloaded graft material. Referring to FIGS. **15A-15B**, the threaded portion of the fixation device may comprise spikes or teeth **80**, which serve to collect native bone or other patient material for distribution through one or more pores located about the circumference of the fixation device. In this embodiment, as the fixation device is advanced into patient's bone, the teeth **80** cut into and shave off portions of the boney anatomy. The shape of the teeth **80** and their proximity to the pores result in the shaved off portions of the patient's bone becoming deposited in the fixation device, which in turn assists with integration between the fixation device and the native bone. Although FIGS. **15A-15B** depict a

generally rectangular pore shape, the shape may be circular, oval, elliptical or other shapes without departing from the novel aspects of the invention.

[0073] In other embodiments the spikes or teeth **80** may be positioned on a device separable from the fixation device, such as the washer of FIG. **16A**. The washer is preferably configured to be coupled to the shaft of the fixation device by a threaded connection, so that the washer may be advanced along the length of the fixation device as shown in FIG. **16B**. The washer of FIGS. **16A-16B** improves fixation in the underlying bone, in part by its advancement along the length of the fixation device for placement and tightening against the patient's boney anatomy, which in turn improves fixation at the entry point and distributes loading forces across a larger surface area than a fixation device without a washer. In embodiments, the washer may contain one or more lateral cannulae through the body of the washer for placement of fixation pins to prevent the washer from moving or backing out while the fixation device is being implanted.

[0074] In yet another embodiment, the washer may be advanced deeper into the bone or countersunk, rather than being positioned on the boney surface, as shown in FIGS. **17A-C**. This embodiment improves fixation and distribution of loading forces. The washer described herein is preferably manufactured from a material that helps absorb forces to prevent stress concentrations. The material of the washer may include, for example, stainless steel, titanium alloy, aluminum alloy, chromium alloy, and other metals or metal alloys.

[0075] The implant length and diameter may be pre-surgically planned to match the anatomical size of the patient's anatomy. The implant porosity and subsequent modulus may be pre-surgically planned to match the bone density of the intended patient. For example, in one embodiment, the surgical devices described above may be matched to an anatomic feature of a patient that has degenerated and needs to be restored. In another embodiment, the surgical device may be necessary to correct structural or physiological deformities present in the patient anatomy, and thereby serve to correct position or alignment of the patient anatomy. Other devices may be patient specific but do not serve a restorative or "structural" function.

[0076] The fenestrations **40** may extend the entire width of the screw for substantially the entire length of the shaft **15**, as shown. In this manner, the number of fenestrations **40** may be increased, thereby increasing the number of locations along the outer surface of the screw that either material can be introduced (via the screw) or material may be collected (via suction as described above). In embodiments, the screw may have a longitudinal channel **35** that runs lengthwise through the screw and permits material to be introduced through the longitudinal channel and into the plurality of fenestrations **40**. For example, the longitudinal channel may be accessible from the head portion **10** of the screw, such that an operator may inject fluid or solid material into the longitudinal channel **35** and ultimately through the porous structure of the screw via fenestrations **40**. In a similar manner, the fenestrations **40** (which surround the outer surface of the shaft **15** of the screw) may be used with vacuum or suction applied to the head portion **10** of the screw to retract material adjacent to the fenestrations **40** and either retained by the porous structure of the screw or within the longitudinal channel **35**. By way of example, the screw could either be infused with bone morphogenetic protein or equivalent to enhance fusion between the screw and the patient's anatomy, or the screw could suction and retain autogenous blood into the fenestrations, which in turn stimulates boney ingrowth in the porous section of the screw. Variations on this embodiment are considered within the scope of the present disclosure.

[0077] Referring to FIGS. **18A-B**, another embodiment of the present disclosure is shown. Here, the screw body **10** and/or longitudinal channel **35** may comprise additional fenestrations **25** in certain portions to better accommodate a particular patient's anatomy. For instance, device may comprise fenestrations **25** along a first and/or second axial surface of the screw **10** (as shown in FIG. **18A**), or may comprise fenestrations that pass through the screw **10** and intersect the longitudinal channel **35** (as shown in FIG. **18B**). The screw **10** and longitudinal channel **35** may comprise a greater number of fenestrations than depicted in FIGS. **18A-B**.

[0078] According to other embodiments, the screw may comprise a varying array of porous sections, portions or components, which may increase or decrease along the length of the screw, including by way of illustrate but not limitation, along a gradient. For example, the density of the porous portions may vary along a portion of the length of the screw, such that there is a decreased density or pattern of porosity along the length of the screw.

[0079] Referring now to FIG. 19, the screw may comprise a modified tip 90, as shown. In embodiments, a diameter of the modified tip 90 may be larger than the pilot hole to insert the screw, while in other embodiments the diameter of the modified tip 90 may be smaller than the pilot hole. The screw may further comprise a modified porosity profile or gradient 40" such that the distal end closest to tip 90 has a greater porosity than at the midsection of screw. In addition, one or more fenestrations 25 may be present along the portion of the screw containing the porous gradient 40". The screw may comprise a longitudinal channel 35" as described above.

[0080] In one of more sections, the screw may not comprise any porosity at all. In this manner, material may either be collected or introduced via the screw only at desired locations along the length of the screw, which may be desirable given the particular bone density and surrounding anatomical features of a particular patient.

[0081] FIGS. 20A-20B depict a screw according to yet another embodiment. In this embodiment, the screw may be customized to match a specific patient by incorporating patient-imaging data into the configuration of the screw, including the imaging data types described above. For example, the thread 30 diameter and/or thickness may be designed according to the patient's specific bone density and location of cortical and cancellous bone in the patient's anatomy. As another example, the longitudinal channel 35" may be adjusted by diameter, and may have a tapering diameter in the direction of either the distal or proximal end of the longitudinal channel 35" depending on the patient's anatomical constraints. In this manner, the longitudinal channel may be matched to achieve a specific fit and/or fill (of the longitudinal channel 35") of cancellous bone according to a specific patient's need, which in turn facilitates endosteal contact and may reduce subsidence after implantation. Referring specifically to FIG. 20B, the diameter of the longitudinal channel 35" may vary without varying the diameter of the shaft of the screw. Likewise, the porosity gradient 40" may vary from one screw to the next to achieve the best results for the specific patient.

[0082] FIGS. 21A-21B show the screw according to another embodiment. Here, the cross-sectional shape of the longitudinal channel 35" may vary from what has previously been described and depicted. For instance, the longitudinal channel 35" may be oval, elliptical, polygonal or otherwise variable in shape from one cross-section from another cross-section of the longitudinal channel 35". Referring to FIG. 21A, the longitudinal channel 35" has a wide diameter with an increase along the length of the screw from the proximate to the distal end. In FIG. 21B, the same screw may comprise a constant diameter about a cross-section taken 90 degrees from the cross-section shown in FIG. 21A. While, in general, the porosity gradient 35" is consistent among FIGS. 21A-B, it is expressly understood that different gradients than the one shown may be incorporated with this embodiment.

[0083] FIG. 22 shows a detailed view of the screw according to an alternate embodiment. Here, the screw may comprise both teeth 80 and channels 85 along the threaded portion of the screw. In certain embodiments, the screw may comprise channels 85 along the threads 30. In embodiments, the screw may comprise channels 85 along both the body and the threads 30 of the screw.

[0084] Further variations of the embodiments are also described. For example, the fixation device may be a pedicle screw, which may be additively manufactured such that the solid and porous aspects of the screw are fused together into a single solid construct. In other embodiments, the solid and porous structure may be co-extensive with the porous elements interdigitated within and around various solid elements of the screw. Variations on these embodiments are considered within the spirit of the presently claimed invention.

[0085] The porous elements of the fixation device, screw or implant may be designed to more

closely resemble that of a specific patient's anatomy. Accordingly, one advantage of the present disclosure is to promote bony ingrowth throughout the porous portions or sections of the implant, which in turn reduces the risk of loosening of the device under dynamic loading situations.

[0086] In still other embodiments, only a portion of the screw is manufactured with a porous surface. For example, the exposed porous aspects of the screw may be localized along the minor diameter of the thread form. The screw may therefore comprise hollow, porous or solid core elements to allow for varying levels of implant stiffness. These areas may be surrounded by a mostly solid thread form to facilitate smooth implantation and firm seating of the screw. Particular reference is made to the Appendix when referring to this embodiment.

[0087] As referred to above, the porous elements of the screw may be designed with localized fenestrations, which may be at least partially accessible from the screw head, and thereby facilitate delivery of osteogenic agents such as bone-morphogenetic proteins, HA and or allograft or autograph tissue into the porous portions of the screw. This in turn allows for bony ingrowth and greater pullout resistance, as described above. The exposed porous structure may be located on the proximal portion of the screw, adjacent the screw head, in order to localize ingrowth. Localization of ingrowth may increase mechanical characteristics of the bone screw interface, and subsequently allow for easier implant removal in the case of revision surgery. The localized porous elements may be tapered outward to increase the interference fit of the porous elements with the surrounding anatomy.

[0088] The porous features are representative of porous cancellous bone with porosity preferably ranging between 10-90% and most preferably between 30-80% to allow for ingrowth of osteocytes and supporting vasculature. Stated another way, the porous features, when compared to the solid features of the device, make up about 30-80% of the volume of the device. In a most preferred embodiment, the porosity is about 45-65%.

[0089] In certain embodiments, the porous structure may be regular and geometric or irregular in form. In yet other embodiments, the porous density may be homogenous throughout the screw, or may be heterogeneous in order to attain desired stiffness and or improve the structural interface of the solid and porous elements. Various additional embodiments are shown in the Appendix comprising a porosity gradient about, by way of example and not limitation, a major dimension of the fixation device.

[0090] An exemplary porosity gradient may be present along the length of a fixation device, such as a screw. In this particular embodiment, the distal end of the screw has a greater porosity than the proximal end. However, in certain embodiments the fixation device may have a greater porosity towards the central axis of the screw, or vice versa, such that the gradient is oriented primarily along the minor axis of the fixation device. In certain embodiments, including those described below, the fixation device comprises a porosity gradient that varies by more than one axis or dimension of the fixation device.

[0091] In embodiments, the gradient or pattern of porosity present in the fixation device may be derived from, at least in part, the properties of a specific patient who receives the fixation device. For example, the gradient may be determined from properties of a patient's bony anatomy, including data obtained from a bone density scanner or equivalent device. Additional data may be obtained from complementary equipment, including but not limited to magnetic resonance imaging (MRI) data, computed tomography (CT) data, x-ray imaging data, bi-planar x-ray imaging data, bone densitometry scan data, medical imaging data, fluoroscopy data, sampled bone material harvested from the patient, or other anatomical data.

[0092] The gradient shown in the Appendix may comprise a denser porous structure towards the distal end, and preferably comprises a gradual reduction in porosity approaching the midline of the fixation device, as shown. In other embodiments, the porous portions of the fixation device may continue throughout the threaded portion of the screw, or may only continue for a very short length of the screw. The fixation device comprising the porous gradient, while preferable for spinal

surgical procedures and especially pedicle screw trajectories, may also be beneficial in other applications, including but not limited to general orthopedic procedures, S1AI and/or S2AI screw trajectories, spinal tethering anchors, suturing anchors and anchors for ACL repair.

[0093] In other embodiments, the screw preferably comprises a solid or non-porous core, which provides adequate strength for the intended application(s). The screw's core, however, may be tapered and decrease the solid or non-porous diameter of the screw as the gradient of porosity becomes more dense (i.e., approaching the distal end of the screw). In other embodiments, this gradient migration is reversed about the major axis of the screw. In other embodiments, the change in diameter of the non-porous core of the screw occurs more gradually over the length of the threaded portion of the screw, and may continue through the entire length of the threaded portion of the screw.

[0094] In embodiments, at least a portion of the distal end of the fixation device may be cannulated or otherwise permit packing of material within at least a portion of the fixation device. This cannulation facilitates preloading the fixation device with autograft, allograft, bioactive or drug eluding material prior to implantation. Alternatively, a user may preload the foregoing material via the pilot hole. In these manners, the fixation device is capable of being preloaded with graft or similar material from within the fixation device, thereby facilitating and promoting boney ingrowth.

[0095] In an alternative embodiment, the cannulation may enable collection of native bone material as the fixation device is being inserted, thereby augmenting or replacing preloaded graft material. For example, the threaded portion of the fixation device may comprise spikes or teeth, which serve to collect native bone or other patient material for distribution through one or more pores located about the circumference of the fixation device. In this embodiment, as the fixation device is advanced into patient's bone, the teeth cut into and shave off portions of the boney anatomy. The shape of the teeth and their proximity to the pores result in the shaved off portions of the patient's bone becoming deposited in the fixation device, which in turn assists with integration between the fixation device and the native bone. Although certain drawings depict a rectangular pore shape, the shape may be circular, oval, elliptical or other shapes without departing from the novel aspects of the invention.

[0096] In other embodiments the spikes or teeth may be positioned on a device separable from the fixation device, such as a washer. The washer is preferably configured to be coupled to the shaft of the fixation device by a threaded connection, so that the washer may be advanced along the length of the fixation device. The washer improves fixation in the underlying bone, in part by its advancement along the length of the fixation device for placement and tightening against the patient's boney anatomy, which in turn improves fixation at the entry point and distributes loading forces across a larger surface area than a fixation device without a washer. In embodiments, the washer may contain one or more lateral cannulae through the body of the washer for placement of fixation pins to prevent the washer from moving or backing out while the fixation device is being implanted.

[0097] In yet another embodiment, the washer may be advanced deeper into the bone or countersunk, rather than being positioned on the boney surface. This embodiment improves fixation and distribution of loading forces. The washer described herein is preferably manufactured from a material that helps absorb forces to prevent stress concentrations. The material of the washer may include, for example, stainless steel, titanium alloy, aluminum alloy, chromium alloy, and other metals or metal alloys.

[0098] The implant length and diameter may be pre-surgically planned to match the anatomical size of the patient's anatomy. The implant porosity and subsequent modulus may be pre-surgically planned to match the bone density of the intended patient. For example, in one embodiment, the surgical devices described above may be matched to an anatomic feature of a patient that has degenerated and needs to be restored. In another embodiment, the surgical device may be necessary

to correct structural or physiological deformities present in the patient anatomy, and thereby serve to correct position or alignment of the patient anatomy. Other devices may be patient specific but do not serve a restorative or “structural” function.

[0099] The surgical devices described herein may be manufactured via additive manufacturing. In the context of spinal implants, the surgical devices may be used in all approaches (anterior, direct lateral, transforaminal, posterior, posterior lateral, direct lateral posterior, etc). Specific features of the surgical device can address certain surgical objectives, for example restoring lordosis, restoring disc height, restoring sagittal or coronal balance, etc. The fixation and surgical devices described herein may then be fabricated by any method. Fabrication methods may comprise the use of a rapid prototyping machine, a 3D printing machine, a stereolithography (STL) machine, selective laser sintering (SLS) machine, or a fused deposition modeling (FDM) machine, direct metal laser sintering (DMLS), electron beam melting (EBM) machine, or other additive manufacturing machine.

[0100] To add further stability to the seating and placement of the fixation devices described herein to the patient anatomy, the outer surfaces of the fixation device may further comprise one or more spikes or teeth or other surface features, which serve to contact and at least partially penetrate or “grip” the patient anatomy to secure the fixation device in place. In one embodiment, the surface features may be made of the same material and may be permanently attached to the fixation device. In another embodiment, the surface features may be comprised of an overlay, and/or may be made of a different material, such as the ones described herein, and may further be selectively inserted onto the fixation device(s) as desired.

[0101] One having skill in the art will appreciate that embodiments of the present disclosure may have various sizes. The sizes of the various elements of embodiments of the present disclosure may be sized based on various factors including, for example, the anatomy of the patient, the person or other device operating with or otherwise using the apparatus, the surgical site location, physical features of the devices and instruments used with the apparatus described herein, including, for example, width, length and thickness, and the size of the surgical apparatus.

[0102] One having skill in the art will appreciate that embodiments of the present disclosure may be constructed of materials known to provide, or predictably manufactured to provide the various aspects of the present disclosure. These materials may include, for example, stainless steel, titanium alloy, aluminum alloy, chromium alloy, and other metals or metal alloys. These materials may also include, for example, PEEK, carbon fiber, ABS plastic, polyurethane, polyethylene, photo-polymers, resins, particularly fiber-encased resinous materials, rubber, latex, synthetic rubber, synthetic materials, polymers, and natural materials.

[0103] One having skill in the art will appreciate that embodiments of the present disclosure may be used in conjunction devices that employ automated or semi-automated manipulation. Various apparatus and implants described herein may be provided to facilitate or control the entry point, angular trajectory, height, and/or head orientation of a screw, for example. This is desirable, particularly with placement of screws in the human body, as it permits a surgeon/user to optimize spinal screw head alignment for subsequent rod insertion across multiple bony landmarks. Additionally, by controlling screw placement, a patient specific rod may be designed and manufactured to either match the pre-planned screw placement, or offer angular corrections in order to optimize curvature of the spine.

[0104] The present disclosure may also be advantageous in light of recent improvements in decentralized manufacturing. For example, surgical devices may soon be capable of fabrication in a number of different and convenient settings, including but not limited to an off-site manufacturing location, an on-site manufacturing location, using equipment present in a surgeon's clinic or offices or in a public or private hospital. For example, modules may be fabricated based on a particular patient need and immediately fabricated once the need is identified, and then provided directly to the surgeon.

[0105] Additional benefits of the systems and methods described herein include improving device fixation, and/or preventing unwanted contact between devices and patient anatomy (e.g. the patient's spinal cord). The further use of methods described above, including the use of software analytics, may further aid in determining screw placement and orientation to achieve the ideal screw placement and/or rod shape. For example, the use of various apparatus described herein to achieve desired screw placement and orientation in turn provides improved alignment of a secondary device, such as a rod, with the screws heads. This benefit in turn allows the surgeon/user to achieve optimal sagittal and/or coronal alignment, which assists in rod placement and improves correction of the patient's anatomy.

[0106] While various embodiments of the present disclosure have been described in detail, it is apparent that modifications and alterations of those embodiments will occur to those skilled in the art. However, it is to be expressly understood that such modifications and alterations are within the scope and spirit of the present disclosure, as set forth in the following claims. For further illustration, the information and materials supplied with the provisional and non-provisional patent applications from which this application claims priority are expressly made a part of this disclosure and incorporated by reference herein in their entirety.

[0107] It is expressly understood that where the term “patient” has been used to describe the various embodiments of the disclosure, the term should not be construed as limiting in any way. For instance, a patient could be either a human patient or an animal patient, and the apparatus and methods described herein apply equally to veterinary science as they would to surgical procedures performed on human anatomy. The apparatus and methods described herein therefore have application beyond surgical procedures used by spinal surgeons, and the concepts may be applied to other types of “patients” and procedures without departing from the spirit of the present disclosure.

[0108] The foregoing discussion of the disclosure has been presented for purposes of illustration and description. The foregoing is not intended to limit the disclosure to the form or forms disclosed herein. In the foregoing Detailed Description for example, various features of the disclosure are grouped together in one or more embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed disclosure requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate preferred embodiment of the disclosure.

[0109] The present inventions, in various embodiments, include components, methods, processes, systems and/or apparatuses substantially as depicted and described herein, including various embodiments, subcombinations, and subsets thereof. Those of skill in the art will understand how to make and use the present inventions after understanding the present disclosure. The present inventions, in various embodiments, include providing devices and processes in the absence of items not depicted and/or described herein or in various embodiments hereof, including in the absence of such items as may have been used in previous devices or processes, e.g., for improving performance, achieving ease and/or reducing cost of implementation.

[0110] Moreover, though the present disclosure has included description of one or more embodiments and certain variations and modifications, other variations and modifications are within the scope of the disclosure, e.g., as may be within the skill and knowledge of those in the art, after understanding the present disclosure. It is intended to obtain rights which include alternative embodiments to the extent permitted, including alternate, interchangeable and/or equivalent structures, functions, ranges or steps to those claimed, whether or not such alternate, interchangeable and/or equivalent structures, functions, ranges or steps are disclosed herein, and without intending to publicly dedicate any patentable subject matter.

Claims

1. A device configured for fixation to a particular anatomical feature comprising: a shaft having a proximal portion and a distal portion, the shaft comprising at least one thread; the shaft further comprising at least a first porous gradient and second porous gradient; the first porous gradient being located towards the distal end of the shaft and containing a first porosity; the second porous gradient being located proximal to the first porous gradient and containing a second porosity; and the at least one thread comprising a thread diameter formed from patient-specific data.
2. The device of claim 1, wherein the density of the first porous gradient is greater than the density of the second porous gradient.
3. The device of claim 1, further composing an internal channel extending substantially the entire length of the shaft
4. The device of claim 3, wherein the at least first and second porous gradients are exposed to the internal channel and configured to receive graft material.
5. The device of claim 3, wherein the internal channel is substantially cylindrical.
6. The device of claim 3, wherein the diameter of the internal channel is variable.
7. The device of claim 3, wherein the internal channel is composed of a first radius oriented about a first axis and a second radius oriented about a second axis, and wherein the first radius is larger than the second radius.
8. The device of claim 1, wherein the depth of the at least first and second porous gradient is variable.
9. The device of claim 3, wherein the proximal portion of the internal channel comprises a first smaller diameter and the distal portion of the internal channel comprises a second larger diameter.
10. The device of claim 9, wherein the second diameter is located proximal to a distal tip of the device and configured to collect biologic material.
11. The device of claim 1, wherein the thread diameter is variable.
12. The device of claim 11, wherein the variability of the thread diameter is determined based patient images selected from the following list: X-ray, MRI, or CT.
13. The device of claim 11, wherein the variability of the thread diameter is determined based on a presurgical implant trajectory.
14. The device of claim 11, wherein the variability of the thread diameter is determined based the location of cortical bone in a patient's boney anatomy.
15. The device of claim 14, wherein the location of cortical bone is determined by patient imaging.
16. The device of claim 1, wherein the shaft diameter is variable.
17. The device of claim 16, wherein the variability of the shaft diameter is determined based patient images selected from the following list: X-ray, MRI, or CT.
18. The device of claim 16, wherein the variability of the shaft diameter is determined based on a presurgical implant trajectory.
19. The device of claim 16, wherein the variability of the shaft diameter is determined based the location of cortical bone in a patient's boney anatomy.
20. The device of claim 19, wherein the location of cortical bone is determined by patient imaging.
21. The device of claim 6, further comprising a marker to position the first and second axis of the device in a specific orientation.
22. The device of claim 1, wherein the device is fabricated by additive manufacturing.
23. The device of claim 1, wherein the at least first and second porous gradient comprise cupping channels configured to collect biological material.
24. The device of claim 1, wherein the at least one thread comprises at least one cupping channel

configured to collect biological material.

25. The device of claim 3, wherein the internal channel is further comprised of a porous matrix.
